

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/ supplementary Winter Examination – 2023

Course: B. Tech.

Branch : Electrical & Allied Engineering

Semester :VII

Subject Code & Name: BTEEOE704B Biomedical Instrumentation

Max Marks: 60

Date: 09-01-24

Duration: 3 Hr.

Instructions to the Students:

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Describe acting potential and resting potential. Illustrate it with help of cardiovascular system in human body.		6
B) Elucidate the biomechanics of bones with necessary schematics. Describe how bones handle mechanical stress to ensure effective movement of body parts.		6
C) Illustrate the construction and working of an ultrasonic transducer with the help of the necessary schematic.		6
Q.2 Solve Any Two of the following.		12
A) Elucidate the manual auscultatory method for measuring blood pressure in detail and briefly explain the types of sphygmomanometer.		6
B) Explain in brief following terms: a) Cardiac output b) Heart rate c) Vital capacity		6
C) Give a brief explanation of how to measure blood pH using a neat diagram.		6
Q.3 Solve Any Two of the following.		12
A) Demonstrate the theory of electrodes in biomedical instrumentation and briefly describe the types of electrodes.		6
B) Explain the method for conducting an electrocardiogram with ECG electrodes using a block diagram.		6
C) Derive output voltage equation for differential amplifier. State significance of differential amplifier in biomedical instrumentation.		6
Q.4 Solve Any Two of the following.		12
A) Demonstrate how computerized tomography can be utilized to diagnose abnormalities in the human body through a clear diagram.		6
B) Describe endoscopic method with neat schematic. Discuss its significance in diagnosis and treatment for human body.		6

- C) Compare different medical imaging modalities, such as X-ray, Ultrasonography and MRI. 6

Q. 5 Solve Any Two of the following. 12

- A) Explain the working of ventilator with help of functional diagram of positive pressure ventilator. 6
- B) Write short note on following terms: 6
- a) Nano robots
 - b) Robotic surgery
- C) Describe the advanced surgical techniques used by medical professionals worldwide. Explain in brief its significance for the current healthcare system. 6

***** End *****